



Thin-Airfoil Theory Solver for NACA 4-Digit Airfoils

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Abstract

This document presents a comprehensive implementation of thin airfoil theory for the NACA four-digit airfoil family. The theoretical foundation is established through the solution of the fundamental integral equation, which ensures that the camber line is a streamline while satisfying the Kutta condition at the trailing edge. The vortex strength distribution is expressed as a Fourier series, with coefficients determined analytically for the NACA four-digit camber geometry. The induced velocity field is computed via adaptive numerical quadrature of the Biot-Savart integrals, enabling the calculation of pressure coefficient distributions and streamline patterns. Results are presented for two representative airfoils: the symmetric NACA 0012 and the cambered NACA 2412 at $\alpha = 5^\circ$. The lift coefficients obtained ($C_l = 0.5483$ for NACA 0012 and $C_l = 0.7761$ for NACA 2412) agree with experimental results, validating the implementation. The pressure and streamline visualizations illustrate the physical effects of camber on the flow field, demonstrating the utility of the solver for educational and preliminary design applications.

1 Introduction

This report presents the implementation of a computational solver based on thin airfoil theory for the analysis of NACA four-digit airfoils. The program is designed to calculate the pressure field, the streamlines around a given airfoil, and the coefficient of Lift, providing both quantitative results and visual insights into the flow physics.

The theoretical foundation is established by solving the fundamental integral equation of thin airfoil theory, which ensures that the camber line behaves as a streamline while satisfying the Kutta condition at the trailing edge. The vortex strength distribution is expressed as a Fourier series, with coefficients determined analytically for the NACA four-digit camber geometry. The induced velocity field is then computed using adaptive numerical quadrature of the Biot-Savart integrals, enabling the calculation of pressure coefficient distributions and streamline patterns.

Two representative airfoils are analyzed to validate the implementation: the symmetric NACA 0012 and the cambered NACA 2412, both at an angle of attack of $\alpha = 5^\circ$. Results are compared against experimental data to assess the accuracy of the program.

This document is organized as follows. Section 2 presents the theoretical background of thin airfoil theory. Section 3 describes the geometry of the NACA four-digit camber line. Section 4 details the analytical computation of the Fourier coefficients. Section 5 outlines the numerical methods used to compute the induced velocity field. Section 6 presents the results and visualizations, and Section 7 provides concluding remarks.

2 Theoretical Background

2.1 Fundamental Equation

For thin airfoils, the flow is modeled as a vortex sheet distributed along the camber line. The objective is to determine the circulation distribution $\gamma(s)$ such that: The camber line

becomes a streamline of the flow, and the Kutta condition is satisfied at the trailing edge. Once $\gamma(s)$ is known, the total circulation Γ and subsequently the lift per unit span L' are obtained through integration.

For a thin airfoil, the vortex sheet lies along the camber line when viewed from moderate distances. From far away, the camber line effectively coincides with the chord line. Thus, the vortex sheet is placed on the x -axis (chord line), while the streamline condition is imposed on the camber line.

For the camber line to be a streamline, the normal velocity component of the freestream $V_{\infty,n}$ must be equal and opposite to the normal velocity induced by the vortex sheet $w'(s)$:

$$V_{\infty,n} + w'(s) = 0 \quad (1)$$

At any point P on the camber line with local slope dy/dx , the normal component of the freestream velocity is:

$$V_{\infty,n} = V_{\infty} \sin \left[\alpha + \tan^{-1} \left(-\frac{dy}{dx} \right) \right] \quad (2)$$

For small angles, this linearizes to:

$$V_{\infty,n} \approx V_{\infty} \left(\alpha - \frac{dy}{dx} \right) \quad (3)$$

2.2 Induced Velocity from the Vortex Sheet

The velocity induced normal to the chord line by a small element of the vortex sheet of strength $\gamma(\xi) d\xi$ located at ξ is:

$$dV_{\text{vortex}} = \frac{\gamma(\xi) d\xi}{2\pi(x - \xi)} \quad (4)$$

The normal component at a point x on the chord line due to the entire sheet is:

$$w(x) = - \int_0^c \frac{\gamma(\xi)}{2\pi(x - \xi)} d\xi \approx w'(s) \quad (5)$$

Combining these results yields the fundamental integral equation of thin airfoil theory:

$$\boxed{\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi)}{x - \xi} d\xi = V_{\infty} \left(\alpha - \frac{dy}{dx} \right)} \quad (6)$$

This equation ensures the camber line is a streamline.

2.3 Solution of the Integral Equation

To solve the integral equation (6), the change of variables:

$$\xi = \frac{c}{2}(1 - \cos \theta), \quad x = \frac{c}{2}(1 - \cos \theta_0) \quad (7)$$

with $\theta, \theta_0 \in [0, \pi]$ yields:

$$\frac{1}{2\pi} \int_0^\pi \frac{\gamma(\theta) \sin \theta}{\cos \theta - \cos \theta_0} d\theta = V_\infty \left(\alpha - \frac{dy}{dx} \right) \quad (8)$$

The solution is expressed as a Fourier series:

$$\boxed{\gamma(\theta) = 2V_\infty \left[A_0 \frac{1 + \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right]} \quad (9)$$

This form satisfies the Kutta condition at the trailing edge ($\theta = \pi$) and allows representation of arbitrary camber shapes.

Substituting (9) into (8) and using standard integral identities gives:

$$\frac{dy}{dx} = (\alpha - A_0) + \sum_{n=1}^{\infty} A_n \cos(n\theta_0) \quad (10)$$

This is a cosine Fourier series on $0 \leq \theta_0 \leq \pi$. For a function $f(\theta)$, the Fourier coefficients are:

$$B_0 = \frac{1}{\pi} \int_0^\pi f(\theta) d\theta \quad (11)$$

$$B_n = \frac{2}{\pi} \int_0^\pi f(\theta) \cos(n\theta) d\theta \quad (12)$$

Comparing with (10), we identify:

$$A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{dy}{dx} d\theta_0 \quad (13)$$

$$A_n = \frac{2}{\pi} \int_0^\pi \frac{dy}{dx} \cos(n\theta_0) d\theta_0 \quad (14)$$

2.4 Aerodynamic Coefficients

The total circulation is:

$$\Gamma = cV_\infty \left[\pi A_0 + \frac{\pi}{2} A_1 \right] \quad (15)$$

Using the Kutta-Joukowski theorem:

$$L' = \rho_\infty V_\infty \Gamma = \rho_\infty V_\infty^2 c \left[\pi A_0 + \frac{\pi}{2} A_1 \right] \quad (16)$$

The lift coefficient is:

$$c_L = \frac{L'}{\frac{1}{2}\rho_\infty V_\infty^2 c} = \pi(2A_0 + A_1) \quad (17)$$

Substituting (13) and (14) into (17):

$$c_L = 2\pi \left[\alpha + \frac{1}{\pi} \int_0^\pi \frac{dy}{dx} (\cos \theta_0 - 1) d\theta_0 \right] \quad (18)$$

The lift slope is constant:

$$\frac{dc_L}{d\alpha} = 2\pi \quad (19)$$

This classical result demonstrates that thin airfoil theory predicts a lift slope of 2π per radian, independent of camber.

3 Geometry of the NACA Four-Digit Camber Line

The NACA four-digit airfoil family defines the mean camber line $y_c(x)$ as a piecewise parabolic function. The defining parameters are:

- m : maximum camber (expressed as a fraction of the chord)
- p : chordwise position of maximum camber (expressed as a fraction of the chord)
- c : chord length (typically normalized to $c = 1$)

3.1 Mapping NACA Four-Digit Nomenclature to Geometric Parameters

The NACA four-digit code follows the structure **M P TT**. For the airfoil NACA 2412, the digits break down as follows:

- First digit (2): $m = 2/100 = 0.02$
- Second digit (4): $p = 4/10 = 0.4$
- Last two digits (12): $t = 12/100 = 0.12$ (maximum thickness, not needed for camber calculations)

The chord length c is a separate scaling parameter, typically set to $c = 1$ for theoretical work.

3.2 Piecewise Definition of the Camber Line

For a point along the chord with $0 \leq x \leq c$, the camber height is defined as:

3.2.1 Forward Portion: $0 \leq x \leq pc$

$$y_c(x) = \frac{m}{p^2} \left[2p \frac{x}{c} - \left(\frac{x}{c} \right)^2 \right] \cdot c \quad (20)$$

3.2.2 Aft Portion: $pc \leq x \leq c$

$$y_c(x) = \frac{m}{(1-p)^2} \left(1 - 2p + 2p\frac{x}{c} - \left(\frac{x}{c}\right)^2 \right) \cdot c \quad (21)$$

These parabolas are continuous in both y_c and its first derivative at the junction $x = pc$, with zero height at the leading and trailing edges.

3.3 Derivative of the Camber Line

Introducing the dimensionless coordinate $\bar{x} = x/c$, the slope dy_c/dx is:

3.3.1 Forward Portion: $0 \leq \bar{x} \leq p$

$$\frac{dy_c}{dx} = \frac{2m}{p^2} (p - \bar{x}) \quad (22)$$

3.3.2 Aft Portion: $p \leq \bar{x} \leq 1$

$$\frac{dy_c}{dx} = \frac{2m}{(1-p)^2} (p - \bar{x}) \quad (23)$$

At $\bar{x} = p$, both expressions give $dy_c/dx = 0$, confirming a smooth peak. At the trailing edge ($\bar{x} = 1$), the slope is $dy_c/dx = -2m/(1-p)$, which is non-zero for any cambered airfoil.

3.4 Transformation to the Theta Variable

Using the standard transformation:

$$\bar{x}(\theta) = \frac{1 - \cos \theta}{2}, \quad \theta \in [0, \pi] \quad (24)$$

The transition point $x = pc$ corresponds to:

$$\cos \theta_p = 1 - 2p \quad (25)$$

The slope as a function of θ is:

$$\boxed{\frac{dy}{dx}(\theta) = \begin{cases} \frac{2m}{p^2} \left(p - \frac{1 - \cos \theta}{2} \right), & 0 \leq \theta \leq \theta_p, \\ \frac{2m}{(1-p)^2} \left(p - \frac{1 - \cos \theta}{2} \right), & \theta_p \leq \theta \leq \pi. \end{cases}} \quad (26)$$

This piecewise definition is linear in $\cos \theta$, enabling analytical integration.

4 Analytical Computation of Fourier Coefficients

The Fourier coefficients are defined by:

$$A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{dy}{dx}(\theta) d\theta \quad (27)$$

$$A_n = \frac{2}{\pi} \int_0^\pi \frac{dy}{dx}(\theta) \cos(n\theta) d\theta, \quad n \geq 1 \quad (28)$$

Because the slope function is piecewise and linear in $\cos \theta$, these integrals can be evaluated analytically. The integration is performed separately over $[0, \theta_p]$ and $[\theta_p, \pi]$, with the substitution $\cos \theta_p = 1 - 2p$ applied to express the results purely in terms of m , p , and n .

4.1 Implementation Strategy

For this project, the symbolic mathematics library `SymPy` is employed to:

1. Define the piecewise slope function symbolically.
2. Perform the definite integrals analytically.
3. Simplify the resulting trigonometric expressions.
4. Generate fast numerical functions via `lambdify`.

The coefficients are computed up to $N = 20$. This truncation is justified because the coefficients decay as $1/n^2$ due to the discontinuity in the derivative of the slope, ensuring negligible contribution from higher-order terms.

4.2 Convergence Verification

To validate the truncation, the slope is reconstructed using:

$$\frac{dy}{dx}(\theta) = (\alpha - A_0) + \sum_{n=1}^N A_n \cos(n\theta) \quad (29)$$

The maximum absolute error between the reconstructed and exact slopes is verified to be less than 10^{-6} .

5 Computation of the Induced Velocity Field

5.1 Biot-Savart Law for a Planar Vortex Sheet

The velocity induced by a vortex sheet element of strength $\gamma(\xi) d\xi$ at a field point (x, y) is given by the 2D Biot-Savart law:

$$du = \frac{\gamma(\xi) d\xi}{2\pi} \frac{y}{(x - \xi)^2 + y^2}, \quad dv = -\frac{\gamma(\xi) d\xi}{2\pi} \frac{x - \xi}{(x - \xi)^2 + y^2} \quad (30)$$

Integrating over the chord yields:

$$u_{\text{ind}}(x, y) = \frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) y}{(x - \xi)^2 + y^2} d\xi \quad (31)$$

$$v_{\text{ind}}(x, y) = -\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) (x - \xi)}{(x - \xi)^2 + y^2} d\xi \quad (32)$$

5.2 Transformation to the Theta-Domain

Using $\xi = \frac{c}{2}(1 - \cos \theta)$ and $d\xi = \frac{c}{2} \sin \theta d\theta$:

$$u_{\text{ind}}(x, y) = \frac{1}{2\pi} \int_0^\pi \gamma(\theta) \frac{y}{\left(x - \frac{c}{2}(1 - \cos \theta)\right)^2 + y^2} \frac{c}{2} \sin \theta d\theta \quad (33)$$

$$v_{\text{ind}}(x, y) = -\frac{1}{2\pi} \int_0^\pi \gamma(\theta) \frac{x - \frac{c}{2}(1 - \cos \theta)}{\left(x - \frac{c}{2}(1 - \cos \theta)\right)^2 + y^2} \frac{c}{2} \sin \theta d\theta \quad (34)$$

where $\gamma(\theta)$ is given by (9).

5.3 Numerical Quadrature

The product $\gamma(\theta) \sin \theta$ is finite and smooth over $[0, \pi]$, as:

$$\gamma(\theta) \sin \theta = 2V_\infty \left[A_0(1 + \cos \theta) + \sum_{n=1}^N A_n \sin(n\theta) \sin \theta \right] \quad (35)$$

This regularity enables the use of adaptive Gauss-Kronrod quadrature via `scipy.integrate.quad`, which provides robust and accurate integration.

5.4 Total Velocity and Pressure Coefficient

The total velocity is the sum of the freestream and induced velocity:

$$u_{\text{total}}(x, y) = V_\infty \cos \alpha + u_{\text{ind}}(x, y) \quad (36)$$

$$v_{\text{total}}(x, y) = V_\infty \sin \alpha + v_{\text{ind}}(x, y) \quad (37)$$

The pressure coefficient is computed from the velocity magnitude:

$$C_p(x, y) = 1 - \frac{u_{\text{total}}^2 + v_{\text{total}}^2}{V_{\infty}^2} \quad (38)$$

This expression holds for any point not on the airfoil surface.

6 Results and Visualization

The program was tested on two representative NACA profiles: the symmetric **NACA 0012** ($m = 0$, $p = 0$) and the cambered **NACA 2412** ($m = 0.02$, $p = 0.4$). Both cases used $\alpha = 5^\circ$, $c = 1.0$, and $N = 20$ Fourier terms.

6.1 Lift Coefficient

The lift coefficients predicted by thin airfoil theory are compared with experimental data in Table 1.

Airfoil	Computational (C_l)	Experimental (C_l)
NACA 0012	0.548	0.59 ^[2]
NACA 2412	0.776	0.74 ^[3]

Table 1: Lift coefficients at $\alpha = 5^\circ$.

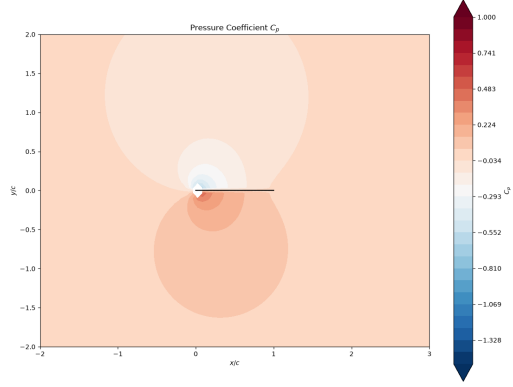
Despite these small differences, the computational results capture the essential trend: the cambered NACA 2412 produces significantly more lift than the symmetric NACA 0012 at the same geometric angle of attack. The camber increases the lift coefficient by approximately 41.5% relative to the symmetric baseline, demonstrating the strong effect of mean camber on aerodynamic performance.

6.2 Pressure Coefficient Field

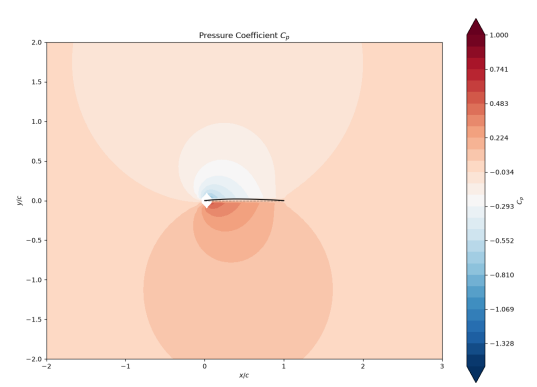
Figure 1 shows the pressure coefficient contours for both airfoils. The leading-edge region is masked to avoid the singularity inherent to the thin-airfoil model.

NACA 0012 (Fig. 1a): The pressure field is almost symmetric about the chord line, which is expected since the symmetry is only broken by the small angle of attack. A stagnation point appears on the lower surface near the leading edge, while a suction peak on the upper surface produces the lowest pressure. The pressure recovers toward the trailing edge, where the Kutta condition ensures a smooth exit.

NACA 2412 (Fig. 1b): Camber shifts the pressure distribution, with stronger upper-surface suction that extends further aft. The lower surface experiences a milder pressure rise. The asymmetry in the pressure field directly corresponds to the higher lift coefficient.



(a) NACA 0012

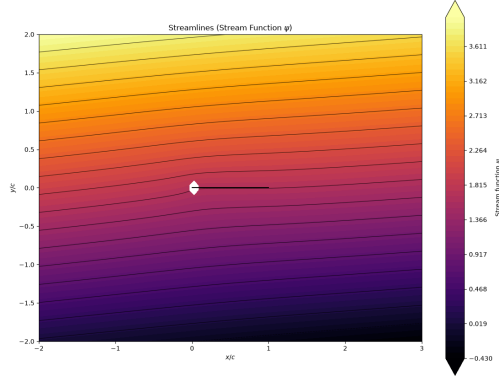


(b) NACA 2412

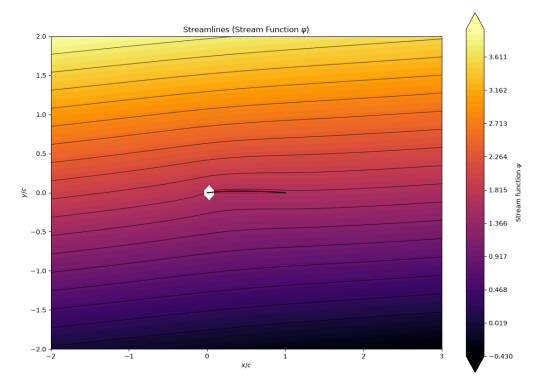
Figure 1: Pressure coefficient fields at $\alpha = 5^\circ$. The solid black curve is the camber line, and the dashed grey line is the chord. Red and blue curves represent the surface pressure on the upper and lower surfaces, respectively.

6.3 Streamline Pattern

Figure 2 presents the stream function ψ as a filled contour plot with overlaid streamline contours.



(a) NACA 0012



(b) NACA 2412

Figure 2: Streamfunction ψ heatmap with streamline contours at $\alpha = 5^\circ$.

The streamlines accelerate over the upper surface and decelerate on the lower surface, consistent with the pressure field. The Kutta condition is naturally enforced: streamlines leave the trailing edge tangentially. For the symmetric case, the streamlines are symmetric about the chord, while for the cambered case they are skewed upward, reflecting the camber-induced circulation.

7 Conclusions

This report has presented a comprehensive implementation of thin airfoil theory for the NACA four-digit airfoil family. The program provides a robust educational tool for understanding the principles of thin airfoil theory and serves as a foundation for more advanced

aerodynamic analyses.

8 References

References

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- [3] Heteyi, C., Molnár, I., & Szlivka, F. (2020). Comparing different CFD software with NACA 2412 airfoil. *Progress in Agricultural Engineering Sciences*, 16(1), 25–40.